

ANDREW Y. KANG

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EDUCATION

- **University of Michigan, Ann Arbor** August 2026 - May 2031 (expected)
Doctor of Philosophy | Electrical and Computer Engineering Ann Arbor, MI
 - **Advisor:** Prof. Inigo Icer
- **Cornell University** August 2023 - May 2026 (graduated 1-year early)
Bachelor of Arts | Double Major in Mathematics and Computer Science Ithaca, NY
 - **GPA:** 4.0/4.0 (4.13/4.3 on Cornell scale)
 - Distinction in All Subjects; *Magna Cum Laude* in Mathematics
 - *Phi Kappa Phi* ($\Phi\K\Phi$)

PUBLICATIONS

- [1] **Andrew Y. Kang**, Yashnil Saha, and Sainyam Galhotra. (2025). **Towards General-Purpose Data Discovery: A Programming Languages Approach**. arXiv Preprint.
- [2] **Andrew Y. Kang**, and Sainyam Galhotra. (2024). **TQL: Towards Type-Driven Data Discovery**. 2024 IEEE International Conference on Big Data (BigData), pp. 7338-7343. IEEE. December 15-18, Washington DC, USA.

PRESENTATIONS

- **Cornell Entrepreneur of the Year: Undergraduate Research Talk** April 2025
 - Represented undergraduate REU research for the college of computing and information science.
 - Orally presented research findings to alumni 'Entrepreneur of the Year' award recipient, John Bicket '02.
- **2024 IEEE International Conference on Big Data: Conference Paper Presentation (Oral)** December 2024
 - Orally presented the paper, *TQL: Towards Type-Driven Data Discovery*.
- **Cornell CIS REU: BURE Symposium (Poster)** August 2024
 - Presented poster summarizing results of summer REU research.

EXPERIENCE

- **Project Lead & Undergraduate Researcher | Advisor: Prof. Sainyam Galhotra** February 2024 - May 2026
Prism Lab | Cornell University Ithaca, NY
 - Studied existing literature in programming languages theory and data systems design.
 - Designed formal syntax and semantics for **TQL**, a novel domain-specific language for data discovery, by leveraging techniques from programming languages and data systems research.
 - Synthesized **ImpRAT**, a foundational algebraic model for data discovery in TQL.
 - Led the architecture implementation by building a prototype system in Python.
 - Exploring and developing potential efficient algorithms for tractable data discovery.
 - Drafted two manuscripts compiling research results.
- **Head Teaching Assistant** August 2024 - May 2026
Courses: Functional Programming, Machine Learning, Formal Verification | Cornell University Ithaca, NY
 - Office Hour Czar : supervised office hour activities & assignments of 45+ TAs.
 - Head Grader : oversaw grading sessions for assignments & exams.
 - Head Proctor : led proctoring team for course exams.
 - Other Duties : led recitation sessions of 40+ students, hosted office hours, and graded assignments & exams.
- **Student Club Founder & President** October 2023 - December 2025
Computational & Quantitative Social Science at Cornell | Cornell University Ithaca, NY
 - Founded and led a student organization interested in building data science and data engineering tools for research in the social sciences.

PROJECTS

- **Compiler for the Eta/Rho Language (Cornell CS 4120/4121 - Compilers)** January 2026 - May 2026
 - Built a complete optimizing compiler targeting x86-64 for the C-like Eta/Rho programming language.
 - Implemented lexer, parser, typechecker and intermediate and assembly code generation.
 - Implemented dataflow analysis and optimization passes (including constant folding, copy propagation, dead-code elimination, and local-value numbering) to improve runtime performance.
 - Collaborated in a group of three members to produce an 18,000 LOC codebase in Java.

RELEVANT COURSEWORK

→ Adv. Linear Algebra (Honors)	→ Real Analysis I & II (Honors)	→ Compilers (incl. Practicum)
→ Abstract Algebra (Honors)	→ Numerical Analysis	→ Systems Programming
→ Kleene Algebra (Grad.)	→ Probability Theory	→ Functional Programming
→ Category Theory (Grad.)	→ Analysis of Algorithms	→ Data Structures & OOP
→ Applied Logic	→ Machine Learning	→ Computer Organization

AWARDS & PRESS

- **Featured Graduate Spotlight for Cornell CIS Class of 2026** *May 2026*
Cornell Bowers College of Computing and Information Science (CIS)
 - Selected by Cornell CIS as the sole representative for the computer science graduates of the Class of 2026.
 - Interviewed and featured across official university websites and social media channels (Linkedin, Instagram, X and Facebook). **Andrew Kang '26: Making the most of undergraduate research opportunities.**
- **2025-2026 CRA Outstanding Undergraduate Researcher Award (Honorable Mention)** *December 2025*
Computing Research Association (CRA)
 - Award recognizes undergraduates across North America who show outstanding potential in computing research.
 - Nominated by Cornell Math Department, received honorable mention from Computing Research Association (CRA) in December 2025.
- **Summer Research Experience for Undergraduates (Grant)** *June 2024 - August 2024*
Cornell Bowers College of Computing and Information Science (CIS)
 - Selected and received \$7000 grant to participate in Cornell's BURE summer REU.
 - Selected and received additional \$1000+ travel grant for conference presentation through Cornell's BURE extension.
- **Cornell CS Department Course Staff Award (Nominee)** *December 2024*
Cornell Bowers College of Computing and Information Science (CIS)
 - Nominated by faculty for course staff award in recognition of excellence in teaching.

SKILLS

Formal Languages: C/C++, Python, Java, Assembly, Rust, OCaml, Rocq, SQL, $\text{\LaTeX}$

Natural Languages: English (native), French (proficient), Mandarin (proficient), Latin (intermediate)

Other Interests: Analytic Philosophy, European History, Soccer Refereeing